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RESEARCH

Navigating Public Sector Transformation: Exploring the Interplay of Digital Leadership, Strategic Agility, and Organizational Performance

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Abstract *This study examines the effect of digital leadership on organizational performance in digital transformation, a strategic priority for public-sector organizations seeking to enhance service quality, transparency, and public value. While digital leadership has been increasingly recognized as a key driver of transformation, less is known about the conditions under which it translates into improved organizational performance in rule-bound public-sector contexts. Drawing on dynamic capability theory, this study examines the role of strategic agility as an enabling organizational capability that shapes the effectiveness of digital leadership. Using survey data from public-sector organizations, this study empirically examines the relationships among digital leadership, strategic agility, and organizational performance. The findings indicate that digital leadership is positively associated with organizational performance; however, this relationship becomes substantially stronger when organizations exhibit higher levels of strategic agility. Specifically, strategic agility enhances organizations' ability to align leadership with intent with coordinated action and timely resource reconfiguration. This study contributes to the literature by clarifying the boundary conditions under which digital leadership yields performance outcomes in public-sector environments characterized by institutional constraints. From a practical perspective, the findings suggest that public-sector digital transformation initiatives should move beyond technology adoption and leadership development alone and, instead, emphasize building strategic agility to ensure effective implementation and sustained performance improvements.*

Keywords Digital leadership · Dynamic capabilities · Fujairah · Organizational flexibility · Organizational performance · Public sector · Strategic agility

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Introduction

In contemporary public-sector environments characterized by regulatory complexity, political oversight, and accelerating technological change, organizational performance increasingly depends on the ability to remain flexible, adaptive, and responsive. From a flexible management perspective, success is no longer driven solely by formal structures or long-term planning; rather, it is driven by organizations' capabilities to sense change, adjust priorities, and reconfigure resources in a timely manner. As public-sector organizations face rising citizen expectations for service quality, transparency, and public value, flexibility has become a central managerial challenge rather than a peripheral capability.

Digital transformation has emerged as a key strategic response to this challenge. However, evidence increasingly suggests that technology alone does not generate flexibility or performance improvements. Instead, the organizational impact of digital initiatives depends on leadership capabilities that align digital investments with strategic priorities, mobilize employees, and enable coordinated action

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across organizational units. Accordingly, digital leadership has gained prominence as a fundamental catalyst of transformation, particularly in environments where adaptation must occur within rigid institutional constraints.

While prior research reports positive associations between digital leadership and organizational performance, most empirical evidence is drawn from private-sector contexts (Daraio et al., 2025; El Sawy et al., 2016; Kane et al., 2019). Public-sector organizations fundamentally differ from private firms because of heightened accountability, regulatory compliance, political scrutiny, and risk aversion, while performance is evaluated through multidimensional public-value outcomes rather than financial metrics alone (Mergel et al., 2019; Moore, 1995). These characteristics often constrain experimentation, slow decision-making, and limit the rapid reallocation of resources. Consequently, leadership capabilities alone may be insufficient to achieve flexible, sustained performance outcomes in public-sector settings.

Drawing on the dynamic capabilities view (Teece, 2018; Teece et al., 1997), this study argues that the effectiveness of digital leadership in public-sector organizations depends on strategic agility. Strategic agility reflects an organization's capability to continuously sense environmental change, align leadership priorities, and flexibly reconfigure resources in response to evolving technological and institutional demands (Clauss et al., 2022; Doz & Kosonen, 2010; Sharma et al., 2025). In rule-bound, bureaucratic environments, strategic agility serves as a core mechanism for enacting flexibility, enabling digital leaders to translate strategic intent into coordinated organizational action despite structural rigidities.

Although prior studies have examined agility as a direct antecedent of organizational performance, limited attention has been paid to its role as a boundary condition that shapes leadership performance relationships, particularly in public-sector contexts. From a management perspective, this represents a critical gap, as the organizational conditions under which leadership-driven digital initiatives yield adaptive and performance-enhancing outcomes remain unclear. Addressing this issue is theoretically important because doing so clarifies when and why flexibility emerges in institutionally constrained organizations.

Accordingly, this study integrates digital leadership and strategic agility within a unified framework and empirically tests the moderating role of strategic agility in the relationship between digital leadership and organizational performance. By focusing on underlying mechanisms rather than geographic novelty, the study contributes to the flexible management literature by demonstrating that flexibility is not an automatic outcome of leadership or technology adoption; rather, it is a contingent capability that is enabled by organizational agility. Practically, the findings suggest that successful public-sector digital transformation requires not only visionary digital leadership but also organizational conditions that support adaptability, timely execution, and effective resource reconfiguration.

Based on this logic, this study addresses two interrelated research questions: whether digital leadership directly enhances organizational performance in public-sector organizations and whether strategic agility strengthens this relationship under conditions of institutional constraints.

Literature Review and Theoretical Foundation

Dynamic Capabilities View as the Theoretical Lens

This study is grounded in the dynamic capabilities view (DCV), which explains how organizations achieve and sustain performance amid rapid change, uncertainty, and environmental turbulence. According to the DCV, competitive and organizational advantage does not stem solely from possessing valuable resources; rather, it also stems from the ability to sense opportunities and threats, seize them through timely strategic decisions, and reconfigure resources accordingly (Teece, 2007; Teece et al., 1997).

The DCV emphasizes that dynamic capabilities are enacted through managerial and organizational processes rather than existing as abstract firm-level attributes. Specifically, Teece (2007) highlights managerial cognition, leadership decision-making, and the orchestration of resources as key microfoundations of dynamic capabilities. This perspective is particularly relevant in public-sector contexts, where rigid structures, accountability pressures, and regulatory constraints limit spontaneous adaptation and require deliberate leadership-driven change.

Within digital transformation settings, the DCV provides a robust theoretical explanation for why some organizations successfully leverage digital technologies while others struggle despite similar technological investments. Digital transformation inherently involves environmental sensing (technological trends and citizen expectations), seizing (strategic prioritization and commitment), and reconfiguring (process redesign, skill development, and resource redeployment). Accordingly, the DCV offers a coherent framework for integrating digital leadership and strategic agility as complementary capabilities that drive organizational performance.

Digital Leadership as a Managerial Microfoundation of Dynamic Capabilities

Drawing on the DCV, digital leadership is conceptualized as a managerial capability that enables organizations to sense digital opportunities, seize them through strategic alignment, and orchestrate digital resources effectively. Unlike traditional leadership, digital leadership integrates technological understanding with strategic vision, change management, and the ability to mobilize employees toward digitally enabled goals (Kane et al., 2019; Sacavém et al., 2025).

From a DCV perspective, digital leaders function as microfoundations that activate dynamic capabilities. They shape how organizations interpret technological signals, prioritize digital initiatives, and coordinate transformation efforts across organizational units (Teece, 2007). Empirical research consistently demonstrates that digital leadership is positively associated with organizational performance, innovation, and adaptability across sectors (Qiao et al., 2024; Wang et al., 2025). However, much of this evidence is drawn from private-sector contexts, leaving public-sector dynamics underexplored.

In this study, digital leadership is conceptualized as a multidimensional construct encompassing distinct yet complementary managerial capabilities that collectively enable digital transformation.

Digital Leadership Capability

Digital leadership capability refers to leaders' proficiency in leveraging digital technologies, data, and virtual communication tools to enhance organizational processes. It enables leaders to sense technological opportunities and guide digital adoption decisions (Mustikasari & Ciptono, 2025; Yang et al., 2023). Within the DCV, these skills represent sensing-oriented managerial routines that support adaptation in digitally turbulent environments.

Digital Leadership Experience

Leadership experience in managing digital initiatives reflects accumulated learning from prior transformation efforts. The DCV emphasizes experiential learning as a critical mechanism through which managers refine their ability to seize opportunities and avoid resource misallocation (Teece, 2007). Leaders with greater digital experience are better positioned to overcome organizational inertia and translate digital investments into performance gains (Lundin et al., 2021; Shin et al., 2023).

Digital Leadership Predictability

Predictive capability captures leaders' ability to anticipate digital trends through analytics, artificial intelligence, and data-driven insights. This dimension directly aligns with the sensing component of the DCV, as predictive analytics enhances foresight and proactive strategic adjustment (Awan et al., 2021). Such abilities are particularly valuable in public-sector organizations facing rapidly evolving service expectations and policy environments.

Digital Leadership Vision

Visionary digital leadership reflects leaders' ability to articulate a compelling digital future and align organizational members around transformation goals. From a DCV perspective, vision facilitates seizing, as it enables coordinated strategic commitment and reduces ambiguity during periods of change (Öngel et al., 2024). In bureaucratic settings, a clear digital vision is essential for legitimizing transformation efforts and sustaining employee engagement.

Strategic Agility as an Organizational-Level Dynamic Capability

While digital leadership provides the managerial impetus for transformation, the DCV suggests that organizational outcomes ultimately depend on a firm's ability to operationalize leadership intent. Strategic agility is an organizational-level dynamic capability that enables continuous adaptation through coordinated sensing, seizing, and reconfiguration (Doz & Kosonen, 2010). Strategic agility is conceptualized as a higher-order capability comprising three interrelated dimensions:

Strategic Sensitivity

Strategic sensitivity reflects an organization's ability to sense environmental changes through continuous scanning, interpretation, and dialogue. This dimension directly aligns with the sensing component of the DCV and enables an early identification of technological, regulatory, and stakeholder-related shifts (de Diego Ruiz et al., 2023).

Leadership Unity

Leadership unity captures the extent to which top management teams share a common strategic understanding and act in a cohesive manner. Within the DCV, leadership unity facilitates seizing, as collective commitment enables timely strategic decisions and reduces coordination failures (Doz & Kosonen, 2010).

Resource Fluidity

Resource fluidity is an organization's ability to rapidly reconfigure and redeploy resources. This dimension corresponds to the reconfiguring component of the DCV and determines whether strategic intent can be translated into action (Alyahya et al., 2023). In public-sector organizations, resource fluidity is often constrained, making it a critical differentiator of performance outcomes.

Integrating Digital Leadership and Strategic Agility within the DCV

Integrating digital leadership and strategic agility within the DCV clarifies how and why leadership influences organizational performance. Digital leadership provides the managerial microfoundations that activate sensing and seizing processes, while strategic agility determines an organization's capacity to execute these processes effectively. Without sufficient strategic agility, digital leadership remains a latent managerial potential that cannot be translated into coordinated organizational action.

Accordingly, this study proposes that digital leadership has a direct positive effect on organizational performance and that strategic agility strengthens this relationship by enabling the effective translation of leadership intent into coordinated organizational action.

Linking Digital Leadership to Agility and Organizational Performance

Digital leadership and strategic agility are increasingly viewed as complementary drivers of organizational performance in digital transformation contexts. Digital leaders enable organizations to interpret technological signals, align stakeholders around a digital agenda, and mobilize coordinated change initiatives, thereby strengthening organizations' ability to sense and seize digital opportunities (Hussein et al., 2024; Kane et al., 2019).

At the organizational level, strategic agility equips firms with the meta-capabilities required to continuously adapt strategies and reconfigure resources in response to environmental turbulence (de Diego Ruiz et al., 2023; Doz & Kosonen, 2010). Empirical evidence indicates that agility-related capabilities enhance digital transformation outcomes by improving responsiveness to stakeholder demands, policy changes, and technological disruption, particularly in public-sector settings (Zhang et al., 2023).

However, existing research often examines digital leadership and strategic agility in isolation and is predominantly grounded in private-sector contexts. This limits understanding of how these capabilities interact within institutionally constrained public-sector organizations, where accountability requirements, regulatory frameworks, and budget rigidity may limit rapid sensing, seizing, and reconfiguration.

Drawing on the dynamic capabilities view, this study argues that digital leadership represents a set of managerial microfoundations that activate sensing and seizing processes, whereas strategic agility constitutes an organizational-level dynamic capability that determines whether leadership intent can be effectively enacted through coordinated decision-making and resource reconfiguration. Accordingly, strategic agility is conceptualized as a boundary condition that strengthens the digital leadership–performance relationship rather than a mediating mechanism that transmits leadership effects sequentially (Al-Husban et al., 2022; Doz & Kosonen, 2010; Mollah et al., 2024).

Recent studies in the flexible management and digital transformation literature have examined related configurations of digital capabilities, agility, and performance across different organizational contexts. For instance, Chaubey (2025) demonstrated that digital transformation capabilities and human resource flexibility enhance organizational responses to environmental uncertainty, particularly when absorptive capacity is high. Ranjan and Narwal (2025) showed that organizational unlearning and agility improve innovation performance during digital transformation in emerging economies. Other complementary streams of research have focused on the evolution of flexible performance management systems under digital transformation (Cosa & Torelli, 2024), the strategic role of coopetition in shaping business performance (Meena et al., 2024), and the alignment between leadership, digital capabilities, and digital strategy in SMEs (Gyamerah et al., 2025).

While these studies collectively highlight the importance of agility, dynamic capabilities, and strategic alignment, they differ from the present study in both focus and theoretical emphasis. Most prior research is grounded in private-sector, SME, or inter-organizational settings and emphasizes operational, learning-based, or relational mechanisms. In contrast, the present study adopts a leadership-centered dynamic capability perspective to explain how digital leadership interacts with strategic agility to shape organizational performance in public-sector organizations characterized by institutional rigidity. Moreover, rather than conceptualizing agility as a direct antecedent or mediating mechanism, this study theorizes strategic agility as a boundary condition that strengthens the digital leadership–performance relationship. This positioning clarifies the study's distinct contribution to the flexible management literature by specifying when and how leadership-driven digital transformation yields performance outcomes in rule-bound public-sector environments.

To enhance theoretical coherence and clarify how the study constructs are derived from the dynamic capabilities view (DCV), Table 1 maps each construct and its dimensions onto the core components of the DCV, namely, sensing, seizing, and reconfiguring, and it links this logic to the hypothesized relationships. Within the DCV framework, organizational performance reflects value creation through the effective orchestration of sensing, seizing, and reconfiguring capabilities rather than isolated leadership actions.

Conceptual Model and Hypotheses

Building on the theoretical arguments presented in the preceding sections, this study adopts a hypothesis-driven approach to examine the relationships among digital leadership, strategic agility, and organizational performance in public-sector organizations. Consistent with prior research, digital leadership is conceptualized as a multidimensional construct encompassing leadership capabilities, experience, predictive capability, and vision. In addition, strategic agility is theorized as a contextual capability that shapes how leadership influences performance outcomes. Accordingly, the following subsections develop hypotheses concerning (a) the direct effects of digital leadership and its dimensions on organizational performance and (b) the moderating role of strategic agility and its underlying dimensions.

Digital Leadership and Organizational Performance

Digital leadership integrates managerial competencies with digital knowledge to harness technology for strategic objectives. Recent research shows that digital leaders not only implement technology but also articulate a clear digital vision, align the organization behind that vision, and foster a culture of innovation (Hussein et al., 2024). By empowering employees and promoting experimentation, digital leaders enhance productivity, creativity, and organizational performance (Qiao et al., 2024). Empirical evidence indicates that organizations led by competent digital leaders achieve superior financial results and experience faster digital transformation than their peers led by less competent digital leaders (Wang et al., 2025). These insights suggest a direct and positive relationship between digital leadership and organizational performance in the public sector. Public-sector organizations with strong digital leadership exhibit higher levels of service quality, innovation, and operational efficiency than do those with weak digital leadership.

H1 *Digital leadership has a positive influence on organizational performance.*

Capabilities Dimension of Digital Leadership and Organizational Performance Digital leadership capabilities encompass a leader's digital skills (e.g., IT proficiency, virtual communication, and knowledge of emerging technologies) and the ability to integrate these capabilities into managerial practice (Mollah et al., 2024). Research highlights that digital leaders with strong technical and managerial capabilities foster organizational adaptability, creativity, and forward thinking (Hussein et al., 2024). They are adept at building virtual teams, leveraging data analytics, and instituting digital cultures that encourage innovation and collaboration (Gao & Gao, 2024; Qiao et al., 2024). These capabilities enable organizations to

Table 1 Alignment of the study constructs with the dynamic capabilities view (DCV)

DCV Component	DCV meaning	Study construct	Dimension	Related hypothesis
Sensing	Identifying opportunities and threats in turbulent environments (Teece, 2007)	Digital Leadership	Digital Leadership Capability	H1-a
		Digital Leadership	Digital Leadership Predictability	H1-c
		Strategic Agility	Strategic Sensitivity	H2-a
Seizing	Mobilizing commitment and making timely strategic decisions	Digital Leadership	Digital Leadership experience	H1-b
		Digital Leadership	Digital Leadership Vision	H1-d
		Strategic Agility	Leadership Unity	H2-b
Reconfiguring	Reallocating and transforming resources to support new strategies	Strategic Agility	Resource Fluidity	H2-c
Performance Outcomes	Value creation through effective orchestration of dynamic capabilities	Organizational Performance	Financial, Customer, Internal Process, and Learning and Growth	H1 and H2

respond quickly to technological disruptions, redesign processes, and deliver better services. Because public-sector organizations often lag behind private firms in digital skills, leadership capabilities become especially critical for improving performance. Leaders with high digital skills and the ability to integrate technology into organizational routines achieve higher performance than leaders with weaker capabilities.

H1a: *Leadership capabilities positively influence organizational performance.*

Experience Dimension of Digital Leadership and Organizational Performance Beyond technical skills, the experience of digital leaders, including their familiarity with digital transformation and history of leading digital initiatives, can shape organizational outcomes. Studies show that executives in organizations with a strong digital legacy push digital transformation more vigorously and align initiatives across the organization, leading to superior performance (Shin et al., 2023). Evidence from a World Economic Forum survey indicates that firms with high digital maturity and well-defined digital strategies achieve significantly greater revenue growth than those without digital leadership (Wang et al., 2025). Experienced digital leaders understand how to navigate organizational inertia, foster digital mindsets, and ensure that digital investments translate into measurable outcomes. Their expertise helps public-sector organizations avoid the common pitfalls of digital transformation and improve performance. Public-sector organizations led by individuals with extensive digital transformation experience display better performance than those led by less experienced digital leaders.

H1b *Leadership experience positively influences organizational performance.*

Predictive Capability Dimension of Digital Leadership and Organizational Performance The growing availability of big data and artificial intelligence tools enables leaders to perform predictive analytics, extracting insights to anticipate trends and optimize decision-making. Scholars argue that predictive analytics capabilities enable organizations to cut costs, shorten production cycles, and adjust strategies based on data-driven evidence (Alghamdi & Agag et al., 2023). Research shows that big data analytics powered by AI significantly improves strategic agility and innovation performance. Such capabilities help leaders sense market changes early, allocate resources efficiently, and tailor services to citizen needs. Additionally, intangible resources, such as data analytics expertise, enhance organizational adaptability and decision-making speed. Thus, digital leaders who leverage predictive analytics are likely to improve organizational performance by making proactive and informed decisions. Leaders who effectively use big data analytics and predictive tools achieve better organizational outcomes than those who rely on traditional, reactive approaches do.

H1c *Leadership predictability positively influences organizational performance.*

Vision Dimension of Digital Leadership and Organizational Performance Vision is a distinguishing attribute of digital leaders. To successfully guide digital transformation, leaders must articulate a compelling digital vision and communicate it throughout the organization (Öngel et al., 2024). Studies emphasize that digital leaders require effective digital communication skills, the ability to integrate culture with digital competence, and the capacity to promote a shared vision (Öngel et al., 2024). When leaders "sell" the digital vision internally and externally, they align employees around common goals, foster engagement, and sustain long-term commitment to transformation. Such visionary leadership motivates employees to innovate, thereby enhancing individual creativity and organizational performance (Öngel et al., 2024; Qiao et al., 2024). In the public sector, where bureaucratic inertia may impede change, a clear digital vision is crucial for rallying stakeholders and justifying investments. Public-sector organizations whose leaders fail to communicate a clear, compelling digital vision often find their transformation initiatives are either absent or poorly articulated.

H1d *Leadership vision positively influences organizational performance.*

Strategic Agility as a Moderator of the Relationship Between Digital Leadership and Organizational Performance

Strategic agility refers to an organization's ability to continuously sense changes in its environment, **make** bold strategic choices, and reallocate resources quickly (Chang et al., 2023). It comprises three interrelated meta-capabilities: (a) strategic sensitivity, which is defined as proactive vigilance and awareness of environmental changes coupled with participative internal dialogue; (b) leadership unity, which is the ability of top management to make collective and timely strategic decisions; and (c) resource fluidity which, is the ability to reconfigure and redeploy resources to support new strategies (de Diego Ruiz et al., 2023). These capabilities enable organizations to adapt to turbulence, exploit opportunities and innovate at speed (Chang et al., 2023; Ludviga & Kalvina, 2025). Prior research suggests that strategic agility strengthens the impact of digital initiatives on performance by allowing

organizations to respond swiftly to new information and overcome bureaucratic inertia.

Because digital leadership sets the direction for transformation while strategic agility determines how rapidly and effectively an organization can follow that direction, it is plausible that strategic agility moderates the relationship between digital leadership and organizational performance. When strategic agility is high, the benefits of digital leadership, such as improved innovation, employee empowerment and digital strategy alignment, are more likely to translate into superior performance. Conversely, low levels of agility may constrain the realization of digital leadership's potential (Alyahya et al., 2023). The positive effect of digital leadership on performance is stronger in organizations that exhibit high levels of strategic agility than in those with low agility.

H2 *Strategic agility positively moderates the relationship between digital leadership and organizational performance.*

Strategic Sensitivity as a Moderator Strategic sensitivity reflects an organization's capacity for proactive environmental scanning and dialogue (de Diego Ruiz et al., 2023). When leaders and employees continuously monitor changes and share insights, they can identify digital opportunities and threats early. Research shows that firms with high strategic sensitivity are more responsive to market shifts and achieve better results in turbulent environments (de Diego Ruiz et al., 2023; Ludviga & Kalvina, 2025). In the context of digital leadership, strategic sensitivity may enhance the effect of leaders' digital initiatives by ensuring that the organization is alert to technological advancements, policy changes and citizen needs. This alertness enables the organization to capitalize on the strategic vision set by digital leaders and avoid potential pitfalls. The positive effect of digital leadership on performance is stronger in organizations with high strategic sensitivity than in those with low sensitivity.

H2a *Strategic sensitivity positively moderates the relationship between digital leadership and organizational performance.*

Leadership Unity as a Moderator Leadership unity refers to the collective commitment of the top management team to make bold and rapid strategic decisions (de Diego Ruiz et al., 2023). It emerges from a collaborative and integrative leadership style that encourages shared responsibility. Evidence suggests that when executive teams act in unison, organizations can implement digital strategies more efficiently and overcome resistance. High leadership unity is therefore likely to strengthen the impact of digital leadership: by providing consistent direction and coordinated action, a unified top management team ensures that digital initiatives are executed coherently across departments, thereby enhancing performance. In contrast, fragmented leadership undermines digital efforts and dampens their effect on organizational outcomes. The positive effect of digital leadership on performance is stronger when the top management team exhibits high leadership unity than when unity is low.

H2b *Leadership unity positively moderates the relationship between digital leadership and organizational performance.*

Resource Fluidity as a Moderator Resource fluidity captures an organization's ability to reconfigure and redeploy financial, human and technological resources to support new strategies (de Diego Ruiz et al., 2023). In fast-moving digital environments, the capacity to shift resources quickly is essential for implementing digital innovations and scaling successful pilots. Research indicates that resource fluidity enables organizations to adapt to new opportunities and maintain competitiveness (Alyahya et al., 2023; de Diego Ruiz et al., 2023). When digital leaders propose new technologies or digital processes, resource-fluid organizations are more capable of allocating funding, talent and infrastructure to realize those proposals, thereby magnifying the performance benefits. In contrast, organizations with rigid resource structures may fail to capitalize on digital leadership. The positive effect of digital leadership on performance is stronger in organizations that can rapidly redeploy resources than in those with low resource fluidity.

H2c *Resource fluidity positively moderates the relationship between digital leadership and organizational performance.*

Together, these hypotheses form a research framework in which digital leadership influences organizational performance through multiple dimensions, and strategic agility and its components amplify this relationship. Empirical testing of these hypotheses in public-sector organizations in Fujairah contributes to the literature by clarifying the mechanisms through which digital leadership and agility interact to shape organizational outcomes.

Figure 1 illustrates our hypothesized model, where digital leadership (DL) affects organizational performance (OP) and strategic agility (SA) acts as a moderator of this link.

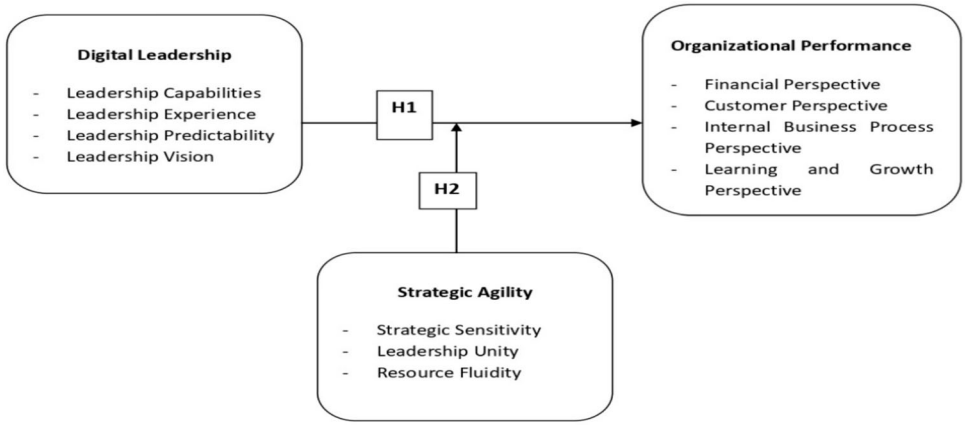

Fig. 1 Conceptual framework linking digital leadership, strategic agility, and organizational performance

Research Design

Research Context and Sample

This study adopts a census-based empirical research design conducted within the public-sector context of the Emirate of Fujairah, United Arab Emirates. Fujairah provides a particularly appropriate empirical setting because of its active engagement in digital government initiatives aligned with the UAE Digital Government Strategy and its sustained investments in digital infrastructure, smart services, and interorganizational digital integration.

Data were collected from three public-sector organizations that collectively represent complementary and strategically important domains of digital transformation within the Emirate: the Fujairah Digital Government Authority, Fujairah International Airport, and the Fujairah Free Zone Authority. These organizations were selected because they reflect diverse operational logics of policy coordination and service digitalization, infrastructure-intensive operations, and digitally enabled trade and logistics while sharing common institutional constraints characteristic of public-sector organizations. This configuration allows for a meaningful empirical investigation of digital leadership and strategic agility within a rule-bound governmental environment.

The target population comprised all managers (senior, middle, and first-line) employed in the three participating organizations. According to official organizational records, the total managerial population consisted of 150 individuals. Given the manageable population size and the study's focus on leadership and strategic decision-making, a census approach was adopted, with all managers invited to participate. This approach eliminated sampling bias and ensured full representation of the managerial population across hierarchical levels.

As a result, complete responses were obtained from all 150 managers, yielding a 100% response rate. The final dataset, therefore, reflects the entire managerial population within the selected organizations rather than a probabilistic sample. This design choice implies that the sample size was determined by population characteristics rather than by arbitrary sampling heuristics.

Notably, commonly cited rules of thumb, such as the "10 respondents per item" guideline, are not universally applicable and have been widely critiqued in methodological research for oversimplifying sample size requirements in multivariate models (Harrington et al., 2013; MacCallum et al., 1999). Simulation studies demonstrate that the required sample size depends on multiple factors, including model complexity, communalities, and the strength of factor loadings, rather than the sheer number of questionnaire items.

Given the theory-driven nature of the proposed model and the use of partial least squares structural equation modeling (PLS-SEM), the sample size is deemed adequate for this study. Prior methodological research indicates that PLS-SEM is particularly suitable for models involving multiple latent constructs and interaction effects and can produce stable and reliable estimates with moderate sample sizes, especially when the population is finite and well defined (Hair et al., 2019b).

Furthermore, the estimated structural model demonstrates strong explanatory power, explaining 75.9% of the variance in organizational performance. This level of explained variance, together with the satisfactory indicator loadings and reliability metrics, is consistent with methodological guidance suggesting that well-fitting models can yield valid and meaningful inferences even with moderate sample sizes (MacCallum et al., 1999).

Table 2 presents the distribution of managerial respondents across the three organizations and hierarchical levels. The data indicate that first-line managers constitute the largest proportion of respondents (60%), ensuring a strong representation of operational perspectives, while senior managers account for 12% of the sample, providing insight

Table 2 Managerial staff by organization and hierarchy level (N = 150)

Management level	Fujairah international airport	Digital government authority	Fujairah Free zone authority	Total	Percentage (%)
Senior management	4	8	6	18	12
Middle management	10	19	13	42	28
First-line management	22	41	27	90	60
Total	36	68	46	150	100

into strategic-level decision-making. This balanced representation strengthens the study's ability to capture leadership practices and strategic agility across organizational levels.

Data Collection Procedure

Data were collected via a structured online questionnaire administered between June and July 2025. The survey was distributed electronically to all 150 managers across the three participating public-sector organizations. Participation was anonymous, and respondents were assured that their responses would be used exclusively for academic research.

To maximize data completeness and coverage, the questionnaire distribution was coordinated with an official digital transformation training program. While attendance at the training was organizationally required, survey participation was explicitly communicated as confidential and non-evaluative. The respondents were informed that their answers would not be linked to individual performance assessments or training outcomes. This procedure resulted in a complete dataset ($N = 150$), consistent with prior public-sector research demonstrating that structured survey administration can yield reliable data when anonymity and transparency are ensured.

Measurement of the Variables

Three latent constructs were measured in this study: digital leadership (independent variable), strategic agility (moderating variable), and organizational performance (dependent variable). All constructs were operationalized via multi-item scales adapted from previously validated instruments to ensure measurement reliability and validity. Unless otherwise stated, all items were measured on a five-point Likert scale ranging from 1 ("strongly disagree") to 5 ("strongly agree").

Digital leadership was conceptualized as a multidimensional construct capturing leadership competencies and behaviors that enable and guide digital transformation. Four dimensions were measured. Digital leadership capability (Items 1–7) captures leaders' ability to effectively leverage digital technologies and tools to improve organizational processes. Digital leadership experience (Items 8–15) reflects leaders' prior involvement in and exposure to digital transformation initiatives. Predictive digital leadership capability (Items 16–22) captures leaders' ability to anticipate digital trends and proactively respond to emerging technological developments. Digital leadership vision (Items 23–31) reflects the clarity and communication of leaders' digital vision and their ability to align organizational members around digital transformation goals.

Strategic agility was modeled as a higher-order construct comprising three interrelated dimensions. Strategic sensitivity (Items 32–37) captures an organization's ability to sense and interpret changes in its external environment. Leadership unity (Items 38–44) reflects the extent to which the top management team shares a common strategic understanding and acts cohesively. Resource fluidity (Items 45–49) captures an organization's ability to rapidly reallocate financial, human, and technological resources in support of new strategic initiatives.

Consistent with prior empirical research, strategic agility was specified as a second-order construct, whereby strategic sensitivity, leadership unity, and resource fluidity were modeled as first-order dimensions.

Organizational performance was operationalized via the balanced scorecard (BSC) framework developed by Kaplan and Norton (1992), which integrates both financial and nonfinancial indicators across four perspectives: financial performance (Items 50–53), the customer perspective (Items 54–57), internal process efficiency (Items 58–63), and learning and growth (Items 64–68). The BSC was selected because it aligns performance measurement with strategic objectives and captures public value beyond purely financial outcomes, making it particularly suitable for public-sector organizations undergoing digital transformation (Madsen, 2025). Compared with alternative frameworks such as the EFQM, the performance prism, and ISO standards, the BSC offers a more pragmatic and strategy-oriented approach, avoiding excessive complexity while enabling holistic performance assessment (Martusewicz et al., 2024).

All measurement items were adapted from established scales reported in the prior literature. Minor wording adjustments were made to enhance the contextual relevance to the public-sector environment in Fujairah. Table 3 summarizes the study constructs, their dimensions, the number of items, and the measurement sources.

Table 3 summarizes each study construct, its dimensions, and the source of its measurement items. For instance, the digital leadership scale is drawn from (Ahn et al., 2014; Yang et al., 2023), and the strategic agility subscale is drawn from (Morton et al., 2018; Reed, 2021).

Instrument Validation and Response Quality

All measurement items used in this study were adapted from established and widely cited scales reported in the prior literature. To ensure content validity and contextual relevance to the public-sector environment in Fujairah, the questionnaire underwent a two-stage validation process. First, three academic experts with expertise in digital leadership, strategic management, and public-sector administration reviewed the instrument and provided feedback on item clarity, relevance, and wording. Minor refinements were made based on their recommendations. Second, a pilot study involving 20 managers from the participating organizations was conducted to assess item comprehensibility and interpretability. The pilot results confirmed the clarity of the items, and no substantive modifications were needed.

Exploratory factor analysis (EFA) was conducted as an initial diagnostic step to examine the underlying factor structure of the adapted scales and to assess their dimensional coherence within the new empirical context. Consistent with best practices in scale adaptation and validation, EFA was employed as an exploratory robustness check rather than a definitive validation technique (Boateng et al., 2018). This step helped ensure that the adapted items loaded meaningfully onto their intended constructs before proceeding to confirmatory analysis.

Confirmatory assessment was subsequently performed using partial least squares structural equation modeling (PLS-SEM) by evaluating the measurement model. Construct reliability and validity were assessed via established criteria, including composite reliability, the average

Table 3 Questionnaire constructs and sources

Construct [role]	Dimension	Number of items	Example item	Sources
Digital leadership [independent variable]	Digital leadership capability	7	"Our leader can effectively leverage new digital tools to improve work processes."	Ahn et al. (2014), Wang et al. (2022), Yang et al. (2023)
	Digital leadership experience	8	"Our leader has considerable experience in managing digital transformation initiatives."	Lundin et al. (2016), Lundin et al. (2021)
	Digital leadership predictability	7	"Our leader can anticipate future digital trends that affect our organization."	Awan et al. (2021), Temelkova (2019)
	Digital leadership vision	9	"Our leader articulates a clear vision for how digital technologies will transform our services."	Luthans (2002), Ruvio et al. (2010)
Strategic agility [moderating variable]	Strategic sensitivity	6	"Our organization quickly senses changes in the environment."	Doz and Kosonen (2008), Morton et al. (2018), Reed (2021)
	Leadership unity	7	"Top managers in our organization share a common strategic understanding."	Doz and Kosonen (2008), Morton et al. (2018), Reed (2021)
	Resource fluidity	5	"Our organization can rapidly redeploy resources to new strategic opportunities."	Doz and Kosonen (2008), Morton et al. (2018), Reed (2021)
Organizational performance [dependent variable]	Financial perspective	4	"Our organization achieves its financial targets."	Jansen et al. (2006), Mohammad et al. (2013), Wu and Lu (2012)
	Customer perspective	4	"Our services meet customer expectations."	Jansen et al. (2006), Mohammad et al. (2013), Wu and Lu (2012)
	Internal process perspective	6	"Our internal processes are efficient and effective."	Jansen et al. (2006), Mohammad et al. (2013), Wu and Lu (2012)
	Learning and growth perspective	5	"Our organization invests in employee development and innovation."	Jansen et al. (2006), Mohammad et al. (2013), Wu and Lu (2012)

variance extracted (AVE), and discriminant validity. In line with contemporary methodological recommendations, confirmatory composite analysis within the PLS-SEM framework served as the primary basis for validating the latent constructs and testing the hypothesized relationships.

While methodological best practices ideally recommend conducting exploratory and confirmatory analyses on separate samples, this was not feasible given the census-based design and the finite population of managers. Accordingly, EFA was treated strictly as an exploratory diagnostic tool, whereas all substantive inferences regarding construct validity and hypothesis testing are derived from the PLS-SEM measurement and structural model results. This approach is consistent with prior empirical research employing PLS-SEM in bounded organizational contexts.

Given the single-source, self-reported nature of the data, potential common-method bias was assessed using Harman's single-factor test. The results indicated that no single factor accounted for the majority of the variance. In addition, procedural remedies, including respondent anonymity, confidentiality assurances, and clear communication of the research purpose, were implemented to further mitigate potential common method bias.

Data Analysis Technique

Data analysis was conducted via SPSS and partial least squares structural equation modeling (PLS-SEM). Descriptive statistics were first computed to summarize the respondent characteristics and provide an overview of the dataset. PLS-SEM was selected for its suitability for theory-driven models with multiple latent constructs and interaction effects, as well as its robustness in studies with relatively small and finite populations.

The measurement model was evaluated by assessing internal consistency reliability and convergent validity via Cronbach's alpha, composite reliability, and the average variance extracted (AVE), following established guidelines for PLS-SEM analysis. Discriminant validity was examined via standard criteria prior to hypothesis testing.

After the measurement model's adequacy was established, the structural model was estimated to test the hypothesized relationships. Digital leadership was specified as the exogenous construct, organizational performance as the endogenous construct, and strategic agility as a moderating variable. Moderating effects were examined via the product-indicator approach. Prior to constructing interaction terms, all latent variables were mean-centered to reduce potential multicollinearity.

Bootstrapping with 5,000 resamples was employed to assess the statistical significance of the path coefficients and moderating effects. Hypotheses were considered supported when the path coefficients were statistically significant at conventional levels. Model evaluation followed established reporting standards for PLS-SEM in quantitative management research, ensuring a transparent and robust assessment of both direct and moderating relationships.

Results

Descriptive Analysis

The demographic characteristics of the respondents are summarized in Table 4, which presents details about their gender, education, age, managerial level, and years of experience.

The demographic profile of the 150 respondents shows that the sample is fairly balanced by gender, with 57.3% male and 42.7% female. In terms of education, nearly half hold a bachelor's degree (49.3%), followed by diploma holders (28.7%), while 16.7% have a master's degree, and a smaller proportion hold a PhD (5.3%), indicating that the sample is generally well educated. With respect to age, the largest group falls within the 36–46 age group (40.7%), followed by the 25–35 age group (33.3%), with fewer respondents aged 47–57 years (21.3%) and only 4.7% above 58 years, suggesting a younger to middle-aged workforce. The distribution of positions indicates that the majority of respondents are supervisors (61.33%), followed by managers (26%), with general managers accounting for 12.67% of the sample.

In terms of professional experience, nearly half of the respondents have less than 12 years of experience (46.7%), while 30.7% have 12–22 years, 14.7% have 23–33 years, and only 8% have 34 years or more, reflecting that most participants are at early to middle stages of their careers. Overall, the sample is moderately gender balanced, highly educated, mostly within the working-age group of 25–46 years, and concentrated in the lower-to-middle range of professional experience.

Construct Reliability and Validity

Cronbach's Alpha and Composite Reliability

The reliability of the study constructs was examined via Cronbach's alpha (CA) and composite reliability (CR). These two measures are widely applied in the literature to evaluate internal consistency and construct reliability. According to Hair et al. (2019a), the accepted threshold for both CA and CR should be higher than 0.70. Values above this cutoff point indicate that the items used in the measurement are consistent and reliable.

Table 4 Respondent demographics

Variable		Frequency	Percentage (%)
Gender	Male	86	57.30
	Female	64	42.70
Education	Bachelor's Degree	74	49.30
	Postgraduate Diploma	43	28.70
	Master's Degree	25	16.70
	Doctoral Degree	8	5.30
Age	25–35 years	50	33.30
	36–46 years	61	40.70
	47–57 years	32	21.30
	Above 58 years	7	4.70
Position	General Manager	19	12.67
	Manager	39	26
	Supervisor	92	61.33
Years of Experience	Less than 12 years	70	46.70
	12–22 years	46	30.70
	23–33 years	22	14.70
	34 years and above	12	8
Total		150	100

In this study, the results in Table 5 indicate that Cronbach's alpha values range from 0.725 to 0.947. Such values are well above the recommended threshold and therefore reflect strong internal consistency among the items. Similarly, the composite reliability values range from 0.753 to 0.953. These values also surpass the minimum required level, confirming that the constructs are measured with sufficient accuracy.

Taken together, these findings suggest that all constructs employed in this study achieved adequate reliability. Hence, the measurement model is psychometrically sound, and the constructs provide a solid foundation for further analysis.

Convergent Validity

It was further imperative to assess convergent validity (the degree to which indicators or scales within a construct measure the same construct).

Table 5 presents the Cronbach's alpha, composite reliability, and average variance extracted (AVE) values for each construct. The Cronbach's alpha values (0.725–0.947) and composite reliability values (0.753–0.953) exceed the 0.70 threshold, indicating strong internal consistency. All AVE values are above 0.50, confirming convergent validity.

In this study, convergent validity was evaluated via the AVE. The AVE indicates the proportion of variance in the indicators that is explained by the construct or latent variable (Henseler, 2018). According to Henseler (2018), the minimum threshold for each construct's AVE is 0.5 or higher. Table 5 shows that all constructs exhibit strong internal consistency, with Cronbach's α (0.725–0.947) and composite reliability (0.753–0.953) exceeding the 0.70 threshold. Each AVE value is above 0.50 (0.501–0.652), confirming convergent validity.

Table 5 Construct reliability and convergent validity

Study dimensions	Number of items	Cronbach's α	Composite reliability	AVE
Digital Leadership	31	.940	0.942	0.652
Leadership Capability	7	.860	0.893	0.545
Leadership Experience	8	.856	0.889	0.503
Leadership Predictability	7	.797	0.853	0.501
Leadership Vision	9	.887	0.909	0.530
Strategic Agility	18	.947	0.953	0.534
Strategic Sensitivity	6	.865	0.881	0.516
Leadership Unity	7	.919	0.882	0.610
Resource Fluidity	5	.836	0.874	0.530
Organizational Performance	19	.926	0.937	0.620
Financial Perspective	4	.912	0.932	0.581
Customer Perspective	4	.892	0.910	0.534
Internal Process Perspective	6	.833	0.845	0.541
Learning and Growth Perspective	5	.725	0.753	0.561

Exploratory Factor Analysis

Exploratory factor analysis (EFA) is a statistical technique used to identify the underlying structure or dimensions within a set of observed variables. It is commonly applied in the early stages of research to uncover the latent constructs that explain the patterns of correlations among the measured items (Henseler, 2018). EFA helps researchers reduce data complexity by grouping related variables into factors, thereby revealing how items cluster together based on shared variance. This method is particularly useful in scale development and validation, where the construct's dimensionality is not yet clearly defined. Unlike confirmatory factor analysis (CFA), which tests predefined hypotheses, EFA is data-driven and does not impose a specific factor structure in advance.

Table 6 Exploratory factor analysis of digital leadership

Variable	Item	Loading (Factor 1)
Digital leadership capability	X11	0.556
	X12	0.440
	X13	0.727
	X14	0.663
	X15	0.777
	X16	0.817
	X17	0.694
Digital leadership experience	X21	0.497
	X22	0.671
	X23	0.796
	X24	0.683
	X25	0.685
	X26	0.564
	X27	0.558
	X28	0.574
Digital leadership predictability	X31	0.435
	X32	0.648
	X33	0.778
	X34	0.620
	X35	0.828
	X36	0.424
	X37	0.415
Digital leadership vision	X41	0.780
	X42	0.806
	X43	0.716
	X44	0.803
	X45	0.788
	X46	0.830
	X47	0.477
	X48	0.490
	X49	0.382

To explore the latent structure of digital leadership, an exploratory factor analysis (EFA) was conducted. Table 6 summarizes the factor loadings for the four identified dimensions: digital leadership capability, digital leadership experience, digital leadership predictability, and digital leadership vision. The purpose of this analysis was to examine whether the observed items cluster meaningfully under their respective constructs, thereby supporting the proposed measurement model.

As displayed in Table 6, most standardized loadings exceed the 0.50 threshold recommended by Hair et al. (2019a), indicating acceptable indicator reliability. Several items exhibit loadings above 0.70, confirming strong convergent validity for most indicators. Nonetheless, a few items (e.g., X12, X31, X36, X37, and X49) show weaker loadings below 0.50, suggesting that they may contribute less effectively to their respective constructs and could be reconsidered in future refinements. Overall, the results validate the multidimensional nature of digital leadership and demonstrate satisfactory measurement quality across its four dimensions.

To assess the underlying structure of strategic agility dimensions, an exploratory factor analysis (EFA) was conducted. Table 7 presents the factor loadings for the three constructs: strategic sensitivity, leadership unity, and resource Fluidity. The results reveal that all items loaded strongly onto their respective factors, with loadings exceeding the recommended cutoff value of 0.50 (Hair et al., 2019a). This finding indicates a clear factor structure and supports the dimensional validity of the measurement model.

As shown in Table 7, most items exhibit high factor loadings of 0.70–0.87, indicating strong indicator reliability and convergent validity. Only one item (M35 = 0.468) marginally falls below the acceptable threshold, implying that it may be reconsidered in subsequent refinements of the measurement model. Taken together, the findings confirm that the constructs of strategic sensitivity, leadership unity, and resource Fluidity collectively form a robust and reliable representation of strategic agility.

To further validate the dimensional structure of organizational performance, an exploratory factor analysis (EFA) was conducted. Table 8 presents the factor loadings for the four balanced scorecard perspectives: financial, customer, internal processes, and learning and growth. Analysis was performed to confirm the construct validity and internal consistency of the performance measurement model.

As illustrated in Table 8, the majority of items loaded strongly onto their intended factors, exceeding the 0.50 threshold recommended by Hair et al. (2019a), which indicates satisfactory indicator reliability and convergent validity. The financial and customer perspectives, with loadings ranging from 0.67 to 0.94, exhibit particularly strong measurement quality. The internal process perspective demonstrates moderate yet acceptable loadings (0.57–0.80). However, one item under the learning and growth perspective (Y45 = –0.117) shows a weak negative loading, suggesting it should be excluded in future refinements of the measurement model. Overall, these results confirm the validity and reliability of the organizational performance construct after the low-loading item is accounted for.

Discriminant Validity

Discriminant validity specifies how a given construct is unique compared to other constructs in a model. In this

Table 7 Exploratory factor analysis of strategic agility

Variable	Item	Loading (Factor 1)
Strategic sensitivity	M11	0.694
	M12	0.738
	M13	0.719
	M14	0.809
	M15	0.756
	M16	0.789
Leadership unity	M21	0.754
	M22	0.735
	M23	0.740
	M24	0.801
	M25	0.781
	M26	0.779
	M27	0.828
Resource fluidity	M31	0.776
	M32	0.863
	M33	0.756
	M34	0.873
	M35	0.468

Table 8 Exploratory factor analysis of organizational performance

Variable	Item	Loading (Factor 1)
Financial perspective	Y11	0.800
	Y12	0.888
	Y13	0.883
	Y14	0.847
Customer perspective	Y21	0.674
	Y22	0.754
	Y23	0.940
	Y24	0.930
Internal process perspective	Y31	0.800
	Y32	0.695
	Y33	0.570
	Y34	0.607
	Y35	0.641
	Y36	0.663
Learning and growth perspective	Y41	0.764
	Y42	0.751
	Y43	0.745
	Y44	0.814
	Y45	-0.117

study, discriminant validity was evaluated using the Fornell–Larcker criterion and the HTMT ratio. The Fornell–Lacker criterion suggests that the square root of a construct's AVE must exceed the correlation values of other constructs. According to Bentiez et al. (2020), the HTMT values in a well-fitted model are expected to be reasonably lower than 0.9.

Table 9 demonstrates discriminant validity: each construct's $\sqrt{\text{AVE}}$ (on the diagonal) exceeds its correlations with the other constructs, meaning that the constructs share more variance with their own indicators than with others.

As shown in Table 10, all HTMT values range from 0.536 to 0.582, well below the 0.90 threshold. These findings provide additional evidence that each construct is empirically distinct.

The results provide evidence that the model's discriminant validity is well established. Based on the Fornell–Larcker criterion (Table 5), the square root of the AVE for each construct (DL = 0.831, SA = 0.812, OP = 0.672) is greater than its correlations with other constructs, indicating that each construct shares more variance with its own indicators than with other constructs. Similarly, the HTMT values (Table 6) between the constructs range from 0.536 to 0.582, all of which are comfortably below the recommended threshold of 0.90 (Henseler, 2018). Taken together, these findings confirm that digital leadership (DL), strategic agility (SA), and organizational performance (OP) are empirically distinct constructs, thereby supporting the measurement model's discriminant validity.

Structural Model

Overall Structural Model Fit

To evaluate how well the hypothesized relationships fit the empirical data, several overall structural model fit indices were examined. Table 11 presents the results of the structural model fit, including the chi-square (χ^2), CFI, TLI, RMSEA, SRMR, and Q^2 statistics. These indices collectively provide evidence on the adequacy of the proposed model for reproducing the observed covariance structure and for assessing its predictive relevance.

As shown in Table 11, the proposed structural model demonstrates an excellent overall fit with the observed data. The χ^2/df ratio of 1.65 lies well within the acceptable threshold (≤ 3), indicating a good balance between fit and parsimony (Kline, 2016). Both the CFI (0.96) and TLI (0.95) exceed the recommended cutoff of 0.90 (Bentler, 1990) and meet the more stringent criterion of ≥ 0.95 [56], confirming strong comparative and incremental fit. The RMSEA value of 0.045 reflects a close approximate fit (Steiger, 1990), and the SRMR of 0.068 is below the acceptable limit of 0.08, supporting overall model adequacy. Although the chi-square statistic is significant ($P < 0.001$), this result is typical for large samples and does not indicate model misspecification. Moreover, the Q^2 value of 0.42 indicates substantial predictive relevance

Table 9 Discriminant validity–Fornell–Larcker

Variables	DL	SA	OP
DL	0.831		
SA	0.518	0.812	
OP	0.525	0.479	0.672

Table 10 Discriminant validity–HTMT

Variables	DL	SA	OP
AC			
SA	0.582		
OP	0.564	0.536	

(Hair et al., 2019a). Collectively, these indices confirm that the structural model is both statistically sound and highly predictive, supporting its suitability for hypothesis testing.

Quality Criteria

After ensuring the model's reliability and validity were satisfactory, its predictive accuracy and relevance were evaluated using R^2 and Q^2. R^2 measures the proportion of variance in the endogenous construct explained by the exogenous constructs, while Q^2 assesses the predictive relevance of the model via the blindfolding procedure. Table 12 presents the quality assessment results, including R^2, adjusted R^2, Q^2, f^2, and VIF values.

As shown in Table 12, the exogenous constructs—digital leadership (DL) and strategic agility (SA)—yield VIF values of 1.890 and 1.819, respectively, which are well below the conservative cutoff of 3.3, confirming the absence of multicollinearity. The endogenous construct, organizational performance (OP), has an R^2 value of 0.759, indicating that approximately 75.9% of its variance is explained by DL and SA. The adjusted R^2 value (0.761) further confirms the model's explanatory power after accounting for complexity. The Q^2 value (0.278) exceeds zero, demonstrating satisfactory predictive relevance, while the effect size ($f^2 = 0.195$) indicates a medium influence of the predictor on OP. Finally, all VIF values (DL = 1.890, OP = 1.710, SA = 1.819) fall well below the critical threshold of 5 [1]. Collectively, these results confirm that the model demonstrates strong explanatory power, predictive accuracy, and structural robustness. Table 12 shows that the model explains 75.9% of organizational performance ($R^2 = 0.759$) and that the VIF values (1.71–1.89) are all below 3.3, indicating no multicollinearity.

Table 11 Model fit indices

Measure	Value
Chi-Square (χ^2)	412.56 ($P < 0.001$)
χ^2/df	1.65
CFI	0.96
TLI	0.95
RMSEA	0.045
Q^2 (Predictive Relevance)	0.42
SRMR	0.068

Hypothesis Testing and Discussion

To verify the hypothesized relationships, the model's path coefficients were estimated using partial least squares structural equation modeling (PLS-SEM). This method was adopted as the primary analytical technique because it aligns with the predictive and exploratory objectives of this study and accommodates the multivariate nature of the model, which examines the effects of the independent variables—digital leadership (DL), leadership capability (LC), leadership experience (LE), leadership predictability (LP), and leadership vision (LV)—on the dependent variable, organizational performance (OP), including the indirect effect of DL on OP through strategic agility (SA).

To justify the methodological choice, partial least squares structural equation modeling (PLS-SEM) was employed in this study because the main objective was to explain and predict organizational performance rather than to confirm an established covariance model. PLS-SEM is particularly suitable for prediction-oriented and exploratory research involving complex models and moderate sample sizes (Benítez et al., 2020; Shmueli et al., 2019).

First, the proposed model includes 68 reflective indicators across three multidimensional constructs—digital leadership (31 items), strategic agility (18 items), and organizational performance (19 items)—which benefit from PLS-SEM's ability to efficiently handle complex hierarchical structures (Benítez et al., 2020). ***Second***, PLS-SEM provides higher statistical power and robust parameter estimates with a sample size of 150, even when multivariate normality cannot be guaranteed (Dijkstra & Henseler, 2015; Sarstedt et al., 2016). ***Third***, the study aims to assess both explanatory and predictive relevance (through R^2, Q^2, and PLSpredict), aligning with contemporary recommendations for prediction-oriented SEM (Shmueli et al., 2019). ***Fourth***, a moderation analysis was conducted using the product indicator approach, a well-established and validated procedure within the PLS framework (Aarstedt et al., 2016).

Table 12 Quality criteria

Variables	R^2	Adjusted R^2	Q^2	f^2	VIF
DL	–	–	–	–	1.890
OP	0.759	0.761	0.278	0.195	1.710
SA	–	–	–	–	1.819

In contrast, covariance-based SEM (CB-SEM) is generally preferred for strict theory-testing contexts with large samples and normally distributed data. Since this study seeks to extend theoretical understanding in an applied, predictive, and context-specific setting, PLS-SEM is the most appropriate and methodologically justified approach. The analysis was conducted via SmartPLS software (version 4.1.0.9), following the two-stage procedure recommended by Hair et al. (2010), where the measurement model was first assessed through confirmatory composite analysis (CCA), followed by evaluation of the structural model to test the hypothesized direct, indirect, and moderating relationships among the constructs.

To test the hypothesized relationships, the structural model was evaluated via the bootstrapping procedure in SmartPLS (version 4.1.0.9) with 5000 subsamples, as recommended by Hair et al. (2010). Table 13 presents the path coefficients and corresponding significance levels for all hypotheses.

As shown in Table 13, digital leadership (DL) has a positive and statistically significant effect on organizational performance (OP) ($\beta = 0.281$, $P < 0.001$), supporting H1. This finding confirms that organizations led by effective digital leaders are better able to achieve superior performance through enhanced decision-making, innovation, and adaptability.

Examining the subdimensions of digital leadership reveals that leadership capabilities (H1a: $\beta = 0.114$, $P = 0.001$), leadership experience (H1b: $\beta = 0.085$, $P = 0.031$), leadership predictability (H1c: $\beta = 0.211$, $P = 0.000$), and leadership vision (H1d: $\beta = 0.152$, $P = 0.004$) all exert significant positive effects on organizational performance. Among them, leadership predictability has the strongest effect ($\beta = 0.211$), followed by leadership vision, indicating that leaders' ability to provide reliable, forward-looking, and visionary guidance plays a pivotal role in improving performance outcomes.

Regarding the moderating effects, the results confirm H2 ($\beta = 0.913$, $P < 0.001$), indicating that strategic agility (SA) significantly amplifies the positive relationship between digital leadership and organizational performance. This finding implies that the performance benefits of digital leadership are maximized in organizations with higher agility. Further analysis of the subdimensions reveals that strategic sensitivity (H2a: $\beta = 0.276$, $P = 0.000$), leadership unity (H2b: $\beta = 0.162$, $P = 0.001$), and resource fluidity (H2c: $\beta = 0.188$, $P = 0.000$) all have significant moderating effects, highlighting the importance of being responsive to environmental shifts, maintaining cohesive leadership alignment, and enabling flexible resource allocation.

Taken together, these findings underscore that digital leadership is a key determinant of organizational success and that its influence becomes substantially stronger when it is integrated with strategic agility. This interaction supports the dynamic capabilities perspective, suggesting that organizations capable of sensing, seizing, and reconfiguring resources through agile leadership are more likely to sustain high performance in dynamic environments.

Figure 2 illustrates the results of the structural model analysis obtained through partial least squares structural equation modeling (PLS-SEM). It visually depicts the path coefficients and significance levels among the study constructs, namely, digital leadership, strategic agility, and organizational performance, showing both the direct and moderating relationships.

As shown in Fig. 2, digital leadership has a strong, positive relationship with organizational performance. The moderating effects of strategic agility (and its subdimensions) are also depicted. We summarize these findings by stating that strategic agility significantly amplifies the DL → OP relationship (supporting H2).

Discussion

This study examined how digital leadership influences organizational performance in public-sector organizations and how strategic agility shapes this relationship through the lens of the dynamic capabilities view (DCV). Beyond confirming direct performance effects, the findings provide important insights into how flexibility is enacted in institutionally constrained environments, offering both theoretical and practical contributions to the flexible management literature.

From a theoretical perspective, the results strongly support the core premise of the DCV that organizational performance emerges not from leadership capabilities in isolation but from their interaction with organizational-level dynamic capabilities (Teece, 2007; Teece et al., 1997). The significant relationship between digital leadership and organizational performance confirms that leadership operates as a critical managerial microfoundation. However, the moderating role of strategic agility indicates that leadership-driven digital initiatives yield performance benefits only when organizations can adapt, realign priorities, and reconfigure resources. From a flexible management perspective, this finding suggests that flexibility is not

Table 13 Path Coefficients and hypothesis testing

Hypothesis	Path	Path Coefficient (β)	*P* value	Result
H1	DL → OP	0.281	0.000	Supported
H1a	LC → OP	0.114	0.001	Supported
H1b	LE → OP	0.085	0.031	Supported
H1c	LP → OP	0.211	0.000	Supported
H1d	LV → OP	0.152	0.004	Supported
H2	SA × DL → OP	0.913	0.000	Supported
H2a	SS × DL → OP	0.276	0.000	Supported
H2b	LU × DL → OP	0.162	0.001	Supported
H2c	RF × DL → OP	0.188	0.000	Supported

an inherent outcome of leadership or technology adoption; rather, it is an enacted capability that depends on organizational conditions.

A particularly important contribution of this study lies in identifying resource fluidity as the most influential dimension of strategic agility. This result highlights that flexibility in public-sector organizations is ultimately constrained less by awareness or strategic intent and more by the ability to redeploy financial, human, and technological resources in a timely manner. Even when digital leaders articulate a clear vision and promote innovation, rigid budgeting procedures, staffing regulations, and administrative routines may prevent organizations from responding flexibly to emerging demands. This finding aligns closely with the DCV's emphasis on reconfiguration as the decisive mechanism through which sensing and seizing activities are translated into performance outcomes (Teece, 2007), and it advances flexibility research by empirically demonstrating the centrality of resource mobility in rule-bound contexts.

The finding that digital leadership vision exerted a stronger influence on performance than leadership experience further enriches the discussion on flexibility. This result suggests that in environments characterized by technological uncertainty, a forward-looking orientation and the ability to articulate a coherent digital future may matter more than accumulated experience alone. Visionary digital leadership appears to play a critical role in reducing ambiguity, aligning organizational actors, and enabling coordinated adaptive responses. From a management perspective, this finding highlights vision as a key mechanism for overcoming inertia and maintaining responsiveness in the face of change.

From a practical perspective, the findings underscore that investments in digital leadership development must be accompanied by organizational reforms that enhance flexibility. Strengthening leaders' digital vision, predictive capabilities, and strategic communication skills can improve performance, but these efforts are unlikely to succeed without parallel initiatives that foster strategic agility. Enhancing strategic sensitivity through systematic environmental scanning, promoting leadership unity through collective decision-making, and enabling resource fluidity through flexible budgeting and talent mobility are essential for translating digital leadership into adaptive and sustainable outcomes. Together, these measures can improve service quality, operational efficiency, and organizational learning in public-sector organizations.

In terms of generalizability, the findings should be interpreted in light of the institutional context of this study. The Emirate of Fujairah operates within a nationally coordinated digital transformation strategy that may strengthen the observed relationships among leadership, agility, and performance. In governance environments that are characterized by weaker coordination or more fragmented institutional arrangements, the enactment of flexibility may face additional constraints. Future research is therefore encouraged to replicate the proposed framework across different public-sector systems, countries, and levels of digital maturity to further explore the contextual variations in flexibility mechanisms.

Finally, although all hypothesized relationships were supported, this outcome reflects the theory-driven nature of the model rather than a lack of contribution. Confirming a theoretically grounded framework in a non-Western public-sector context extends the empirical scope of existing theory and strengthens its explanatory relevance. More importantly, the variation in effect magnitudes, particularly the dominant role of resource fluidity, provides new insight into how flexibility is operationalized in public-sector organizations, moving beyond simple hypothesis confirmation toward a deeper understanding of adaptive performance under institutional constraints.

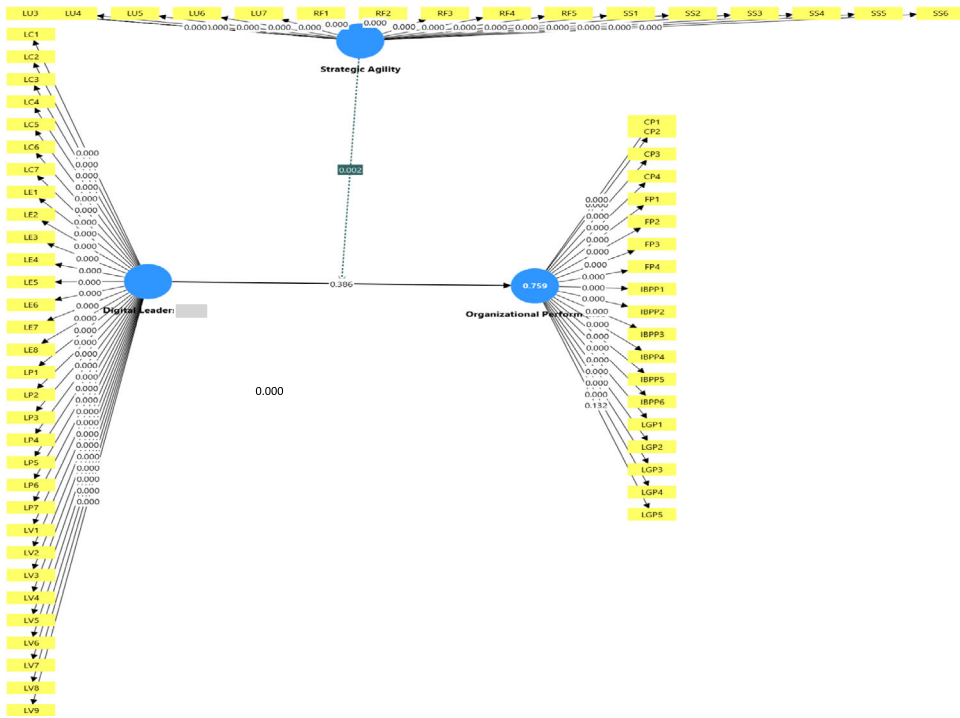

Fig. 2 Structural model path coefficients (PLS-SEM Results)

Implications and Future Research

This study offers several important theoretical contributions to the literature on digital transformation, leadership, and public-sector management. ***First***, the findings demonstrate that digital leadership functions as a dynamic capability in public-sector organizations, enabling agencies to sense technological changes, seize emerging opportunities, and reconfigure resources in response to environmental turbulence. By empirically demonstrating that leadership attributes such as vision, experience, and predictive capability serve as managerial microfoundations for sensing and seizing, this study extends the DCV, which has traditionally been developed in private-sector contexts, to bureaucratic public-sector environments. In doing so, it directly responds to calls for testing the applicability of dynamic capabilities frameworks under conditions of institutional rigidity and public accountability (Mergel et al., 2019). The results confirm that the core logic of sensing, seizing, and transforming remains valid in public administration, thereby reinforcing recent conceptual arguments that position digital leadership as central to the development of digital dynamic capabilities (Albannai et al., 2024).

Second, this research advances theory by clarifying the interaction between digital leadership and strategic agility in driving organizational performance. While prior studies have predominantly conceptualized agility as a mediating mechanism that transmits the effects of leadership or digital capabilities to performance outcomes (Atobishi et al., 2024), the present study provides empirical evidence that strategic agility can also function as a contextual moderator. Specifically, the positive impact of digital leadership on organizational performance becomes significantly stronger under conditions of high strategic agility—characterized by elevated strategic sensitivity, strong leadership unity, and high resource fluidity. This finding introduces a critical theoretical refinement: leadership effectiveness is contingent upon the organizational context. In agile organizations, digital leaders are better positioned to translate strategic intent into coordinated action and measurable outcomes, whereas in rigid settings, the same leadership capabilities yield weaker performance effects. This insight enriches leadership theory by integrating it with research on strategic agility and underscores the importance of examining leadership and organizational capabilities jointly rather than in isolation.

Third, this study contributes valuable empirical evidence from a Middle Eastern public-sector context, addressing a notable gap in the literature. Research on digital leadership and strategic agility has largely focused on Western or private-sector settings, with limited attention to government organizations operating under different administrative cultures and institutional constraints (Mergel et al., 2019). By examining public-sector agencies in the Emirate of Fujairah, this study provides context-

sensitive insights into how local governance structures and policy environments shape digital transformation outcomes. The findings demonstrate that even in traditionally bureaucratic systems, visionary and technologically competent leadership can enhance organizational performance, particularly when it is supported by organizational agility. At the same time, the results highlight that institutional rigidity may dampen agility and thus constrain leadership effectiveness, emphasizing the need to incorporate cultural and administrative context into theories of digital transformation.

In summary, these contributions indicate that this study does not merely replicate established frameworks in a new geographical setting; rather, it refines and extends them by specifying their boundary conditions. By empirically demonstrating that strategic agility moderates rather than mediates the relationship between digital leadership and organizational performance in public-sector organizations, this study introduces a novel theoretical perspective within the dynamic capability literature. This refinement strengthens the research's originality and provides a conceptual bridge between leadership theory and public-sector digital transformation scholarship (Caputo et al., 2025; Ly, 2023).

More broadly, the findings advance a more contextually grounded theory of leadership in the digital era. By integrating dynamic capabilities and strategic agility perspectives, the proposed framework offers a robust analytical lens for examining leadership-driven performance across diverse organizational settings. Accordingly, this study goes beyond testing existing theory in a new domain and creates new avenues for understanding how visionary digital leadership, when coupled with organizational agility, enables effective and sustainable transformation in public-sector environments.

Managerial Implications

The findings of this study offer clear, actionable implications for public-sector leaders and policymakers responsible for steering digital transformation initiatives. By demonstrating that strategic agility significantly amplifies the performance effects of digital leadership, this study highlights that leadership development alone is insufficient unless it is embedded within agile organizational systems. Accordingly, the following implications translate the empirical results into concrete managerial actions aligned with Fujairah's institutional context.

First, leadership development programs in public-sector organizations should be redesigned to integrate digital and behavioral competencies in a balanced manner. Beyond basic digital literacy, leaders require advanced capabilities in strategic foresight, change leadership, and emotional intelligence to guide organizations through complex digital transitions. Prior research suggests that public-sector leaders who combine technological expertise with transformational behaviors—such as trust-building, inclusive communication, and inspiration—are more effective at fostering innovation and performance improvements (AlNyaimi et al., 2022; Sacavém et al., 2025). In hierarchical administrative settings such as Fujairah, strengthening these competencies can reduce siloed decision-making, improve cross-unit coordination, and create the psychological safety necessary for experimentation. Structured initiatives such as mentoring schemes, peer-learning platforms, and targeted digital-transformation bootcamps for middle and senior managers can support this shift.

Second, agility-oriented competencies should be explicitly embedded within leadership training and evaluation systems. Given this study's evidence that strategic agility strengthens the digital leadership–performance relationship, public agencies should prioritize developing leaders' adaptability, problem-solving skills, and cross-functional collaboration capabilities (Atobishi et al., 2024). Scenario-based learning, digital transformation labs, and simulation exercises can enable leaders to practice iterative decision-making, risk assessment, and rapid resource redeployment—the core principles of agile governance. For instance, expanding the curricula of institutions such as the Fujairah Digital Government Academy to include agile project management, stakeholder cocreation, and experimentation-oriented approaches can foster more proactive and resilient leadership mindsets.

Third, organizational governance mechanisms must be redesigned to support agile ways of working. Traditional bureaucratic structures often constrain the translation of digital leadership into performance outcomes by slowing decision-making and limiting coordination. To address this challenge, public-sector organizations should establish cross-functional teams that integrate IT, operations, and policy functions under shared accountability for digital initiatives (Sacavém et al., 2025). Operating through short planning cycles, frequent feedback loops, and iterative reviews, such teams can accelerate implementation while maintaining strategic control. Moreover, the leadership unity identified in this study as a key dimension of strategic agility can be strengthened through executive steering committees that align departments and reduce fragmentation in digital decision-making.

Fourth, resource fluidity should be institutionalized as a core managerial principle to enable the rapid scaling of successful digital initiatives. The findings highlight that

flexibility in reallocating financial, human, and technological resources is critical for converting digital leadership intent into tangible performance gains. Public-sector organizations should therefore introduce flexible budgeting mechanisms, earmarked innovation funds, and policies that allow staff to be temporarily redeployed across digital projects (Atobishi et al., 2024). Encouraging multiskilling and rotational assignments can further enhance organizational adaptability. In the context of Fujairah, where annual budget cycles may constrain responsiveness, pilot-based funding models and staged investment approvals can significantly improve agility without undermining accountability.

Fifth, cultivating an agile organizational culture is essential for sustaining digital transformation. Leaders play a central role in promoting collaboration, continuous learning, and openness to experimentation. Mechanisms such as internal innovation councils, idea-sharing platforms, and recognition systems that reward creative problem-solving can reinforce these cultural attributes. Empirical evidence suggests that when employees perceive strong leadership support for learning and experimentation, their engagement and innovative behaviors increase markedly (AlNyaimi et al., 2022). In Fujairah's public-sector environment, celebrating incremental digital successes and providing controlled "sandbox" environments for testing new technologies can build organizational confidence and accelerate adoption.

Finally, all managerial interventions should remain sensitive to local institutional and cultural conditions. The wholesale transfer of private-sector digital practices often proves ineffective in Gulf-region public-sector organizations because of bureaucratic traditions and distinct governance norms. Therefore, leadership development, agile practices, and governance reforms should be tailored to Fujairah's administrative realities and be aligned with national priorities such as the UAE Digital Government Strategy 2025. Collaboration with regional universities, public innovation labs, and local technology partners can help ensure that digital transformation efforts remain both globally informed and locally grounded.

In summary, the managerial implications of this study move beyond generic prescriptions by demonstrating how digital leadership and strategic agility jointly shape performance outcomes. By investing in integrated digital agility leadership development, redesigning governance structures to enhance flexibility, and contextualizing reforms within Fujairah's institutional environment, public-sector leaders can more effectively translate digital vision into sustained organizational performance.

Limitations and Future Research

Despite the contributions of this study, several limitations should be acknowledged and serve as avenues for future research.

First, this study was conducted within a specific public-sector context in the Emirate of Fujairah. While adopting a census approach ensured full coverage of the accessible managerial population and eliminated sampling bias within the studied organizations, the contextual specificity may limit the generalizability of the findings to other public-sector environments or national settings. Public-sector organizations vary considerably in terms of their governance structures, regulatory frameworks, and levels of digital maturity. Future research is therefore encouraged to replicate the proposed model across different governmental contexts, regions, or countries to enhance external validity and enable comparative analysis.

Second, although the sample size reflects the entire population of managers within the participating agencies (N = 150), it remains moderate in absolute terms, which is common in census-based public-sector studies conducted in bounded organizational settings. While PLS-SEM is well-suited for complex models with moderate sample sizes and demonstrates strong explanatory power in the present study, future research could further strengthen empirical generalization by employing larger samples, multiagency datasets, or cross-national designs.

Third, this study relied on cross-sectional, self-report survey data, which may raise concerns regarding common method variance and causal inference. Although procedural remedies were applied to mitigate these risks, future studies could adopt longitudinal or mixed-method approaches, for example, combining survey data with archival performance indicators or qualitative interviews, to better capture the dynamic evolution of digital leadership and strategic agility over time.

Finally, this study focused on strategic agility as a moderating capability that conditions the effectiveness of digital leadership. While this conceptualization is theoretically grounded in the dynamic capabilities view, future research may explore alternative model configurations, including mediated or multilevel frameworks. Incorporating additional organizational capabilities, such as digital culture, employee engagement, knowledge

integration, or innovation orientation, could provide deeper insight into the mechanisms by which digital leadership translates into sustained organizational performance.

Despite these limitations, this study offers a theoretically grounded and empirically validated framework that advances the understanding of how digital leadership and strategic agility interact to shape performance outcomes in public-sector digital transformation contexts.

Conclusion

This study set out to examine how digital leadership influences organizational performance in public-sector organizations and to clarify the contingent role of strategic agility through the lens of the DCV. Drawing on empirical evidence from public-sector organizations in the Emirate of Fujairah, the findings provide a coherent and theory-driven explanation of why digital leadership succeeds in some contexts while remaining underutilized in others.

The results confirm that digital leadership constitutes a critical managerial microfoundation that positively contributes to organizational performance. However, this study demonstrates that leadership capabilities alone are insufficient to generate sustained performance improvements. Instead, their effectiveness depends on an organization's ability to operationalize leadership intent through strategic agility. This finding aligns with the DCV's core argument that value creation emerges from the orchestration of sensing, seizing, and reconfiguring capabilities rather than from isolated resources or leadership traits.

A key contribution of this study lies in identifying strategic agility as a boundary condition that strengthens the relationship between digital leadership performance. Among its dimensions, resource fluidity emerged as the most influential moderator, underscoring the importance of redeploying financial, human, and technological resources to translate digital vision into tangible outcomes. In highly regulated public-sector environments, this finding underscores the notion that flexibility in resource allocation is not merely an operational concern; rather, it is a strategic capability with direct performance implications.

The relative strength of digital leadership vision compared with that of leadership experience further enriches the theoretical contribution. This result suggests that a forward-looking orientation, clarity of purpose, and the ability to articulate a compelling digital future are particularly critical under conditions of technological uncertainty. In this sense, digital leadership operates less as a function of accumulated experience and more as a strategic capability that shapes organizational sense-making and coordinated action.

From a practical perspective, the findings offer clear guidance for public-sector decision- makers. Investments in digital leadership development should be complemented by organizational reforms that enhance strategic agility. Developing leaders' digital foresight and communication skills, fostering collective alignment on leadership, and enabling flexible resource reconfiguration are necessary conditions for realizing the performance benefits of digital transformation initiatives. The high explanatory power of the proposed model suggests that such integrated investments can substantially improve service quality, operational efficiency, and organizational learning in government institutions.

While this study provides robust insights, its conclusions should be interpreted within the institutional context in which the data were collected. The Emirate of Fujairah operates within a nationally coordinated digital transformation agenda that may amplify the observed relationships. Consequently, the generalizability of the findings to other public-sector settings with different governance structures or levels of digital maturity should be approached with caution. Future research is encouraged to replicate and extend the proposed framework across diverse national, sectoral, and institutional contexts.

Overall, this study advances the understanding of digital leadership in the public sector by demonstrating that leadership effectiveness is contingent upon organizational agility. By integrating digital leadership, strategic agility, and performance within a dynamic capabilities framework, this research provides a theoretically grounded and empirically supported explanation of how public-sector organizations can better navigate digital transformation and achieve sustainable performance outcomes.

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Author Contribution H.A.H.M. = Hossam Ahmed Hanafy Mahmoud M.E.M.M. = Mohamed El-Mohamady El-Mady O.B.A. = Obaid Barman Alzahmi H.A.H.M. developed the research framework, supervised the study, and contributed to writing and revising the manuscript. M.E.M.M. contributed to the theoretical background, data analysis, and interpretation of results. O.B.A. coordinated data collection from government departments and provided contextual insights from the public sector. All authors discussed the results, contributed to the final manuscript, and approved the submitted version.

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Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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Key Questions

1. Which aspects of digital leadership development (vision, experience, or predictive capability) are most critical for enabling adaptive performance in government organizations, and why?
2. How do existing budgeting rules and human resource regulations shape the degree of resource fluidity available to public-sector managers during digital transformation initiatives?
3. To what extent might national culture, administrative traditions, and governance models alter the way in which digital leadership and strategic agility interact to influence organizational performance?
4. Under what organizational or institutional conditions could strategic agility shift from acting as a moderating factor to functioning as a mediating mechanism in digital transformation processes?

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